

Bangladesh University of Business and Technology
Topic: **Introduction to Statistics**

❖ **Definitions of Statistics**

Statistics is a body of methods of obtaining and analyzing data to base decisions on them. It is a branch of scientific methods used in dealing with phenomena that can be described numerically by count or measurement.

As a subject, statistics is concerned with **scientific methods** for collecting, organizing, summarizing, presenting, and analyzing sample data, drawing valid conclusions about population characteristics, and making reasonable decisions based on such analysis.

According to the above definition, there are **five stages** in a statistical investigation:

- i. **Collection:** Data collection constitutes the first step in a statistical investigation. We have to be very careful in collecting data.
- ii. **Organization:** Data collected from published sources are generally in organized form. However, a large mass of figures that are collected from a survey frequently need organization.
- iii. **Presentation:** After the data have been collected and organized they are ready for presentation. Data presented in an orderly manner facilitates statistical analysis.
- iv. **Analysis:** Methods used in analyzing the presented data are numerous ranging from simple observation of data to complicated, sophisticated, and highly mathematical techniques.
- v. **Interpretation:** The last stage is concluding the data collected and analyzed. If the data that have been analyzed are not properly interpreted, the whole object of the investigation may be defeated and fallacious conclusions drawn. Correct interpretation will lead to a valid conclusion of the study and thus aid in decision-making.

❖ **Types of statistics:** There are two types of statistics.

- Descriptive Statistics
- Inferential Statistics

Descriptive statistics- statistics that are used to describe some aspect of the data, such as the typical score or the variability of scores.

Inferential statistics- statistical procedures that are used to conclude the populations under study based on the sample data from those populations.

❖ **Nature of Statistics**

- Statistics must be expressed in numerical form any other form of facts is not statistics
- Statistics deals with the population or aggregate of individuals rather than with individuals.
- Statistical inferences are uncertain.
- Statistics should be affected by a large number of random causes.

❖ **Data:** The raw material of statistics consists of numbers or observations usually obtained by some process of counting or measurement is known as data.

Primary data: Primary data come mainly from direct field operations, which may either be a census or a specially designed survey.

Secondary Data: Secondary data means data that are already available i.e. they refer to the data which have already been collected and analyzed by someone else.

Internal Data: Information, facts, and data useful to the decision-making process available from within a company's information systems is called internal data. Internal data is normally not accessible by outside parties without the company's express permission.

❖ **Information**

Information is data or a collection of data that is (1) accurate and timely, (2) specific and organized for a purpose, (3) presented within a context that gives it meaning and relevance, and (4) can lead to an increase in understanding and decrease in uncertainty. Information is valuable because it can affect behavior, a decision, or an outcome. For example, if a manager is told his/her company's net profit decreased in the past month, he/she may use this information as a reason to cut financial spending for the next month. A piece of information is considered valueless if, after receiving it, things remain unchanged.

Variable:

In statistics, a **variable** has two defining characteristics:

- A variable is an **attribute** that describes a person, place, thing, or idea.
- The value of the variable can "**vary**" from one entity to another.

Example:

- i. The heights of adult males
- ii. The weights of preschool children

❖ Types of Variable:

There are two basic types of variables

1. Qualitative
2. Quantitative

1. Qualitative Variable or an attribute:

When the characteristic being studied is non-numeric, it is called a qualitative variable or an attribute. For this variable numerical measurement is not possible. We can categorize this mutually exclusive data.

Examples:

Qualitative variables are gender, religion, affiliation, type of automobiles owned, state of birth, and eye color.

2. Quantitative Variable:

When the variable studied can be reported numerically, the variable is called a quantitative variable.

Examples:

Quantitative variables are the balance in your checking account, the ages of company presidents, the life of an automobile battery (such as 42 months), and the number of children in a family.

Quantitative variables are either discrete or continuous.

❖ Frequency distribution

When the data set is bigger, to find particular characteristics of a variable we need to calculate the frequency distribution which is introduced as a tabular method of

summarizing data. This frequency distribution can also be presented graphically through several diagrams, charts, and curves.

Formally the frequency distribution can be defined as follows:

A set of classes together with the frequencies of occurrence of values in each class in a given set of data, presented in a tabular form, is referred to as a frequency distribution.

➤ **Formation of a Frequency Distribution**

The process of preparing this type of distribution is straightforward. We have just to count the number of times a particular value is repeated which is called the frequency of that class. To facilitate counting, prepare a column of “tally”. In the other column, place all possible variable values from the lowest to the highest. Then put a bar (vertical line) opposite the particular value to which it relates.

We finally count the number of bars corresponding to each variable value and place it in the frequency column.

Example:

The number of refrigerators sold in 22 working days by a leading agency house:

23 30 20 26 30 20 23 40 40 26 20 30
23 40 28 26 23 40 28 28 30 30

Frequency distribution of the number of refrigerators sold:

No of Refrigerators	Tally	Frequency no of Days
20	III	3
23	IIII	4
26	III	3
28	III	3
30	IIII	5
40	IIII	4

The table clearly shows that on 3 days 20 refrigerators were sold each day, on 4 days 23 refrigerators were sold each day etc.

❖ **Cumulative Frequency:**

In some situations, we may be interested, not in the frequencies in various classes, but rather in the frequencies or proportions of observation which are “less than” or “greater than” a given value. This leads to a cumulative frequency distribution. This is derived from a frequency distribution by forming a cumulative frequency column. This column is computed by adding the successive class frequencies from top to bottom.

The entry corresponding to the top interval is the frequency of that class., the entry opposite the second interval is the sum of the frequencies in first- and second-class intervals, etc., and so on.

If we divide the cumulative frequency by N, the total number of observations, we get the relative cumulative frequencies, which are often expressed in percentages.

Value	Frequency	Cumulative frequency	Relative frequency
0-10	4	4	4/96
10-20	12	16	16/96
20-30	24	40	40/96
30-40	36	76	76/96
40-50	20	96	96/96

❖ **Methods of classifying the data according to class interval:**

There are two methods of classifying the data according to class intervals namely

- a. Exclusive method
- b. Inclusive method

a. Exclusive Method:

When the class intervals are so fixed that the upper limit of one class is the lower limit of the next class it is known as the ‘Exclusive’ method of classification. The following data are classified on the basis:

Income (Tk.)	No of Employees
1800-1900	50
1900-2000	100
2000-2200	200

It is clear that the ‘Exclusive method’ ensures continuity of data in as much as the upper limit of one class is the lower limit of the next class. Thus, in the above example, there are 50 persons whose income is between Tk. 1800 and Tk. 1888.99. A person who is getting exactly Tk. 1900 would be included in the class 1900-2000.

Here, whenever this method is used it is necessary to give clear instructions in the questionnaire. However, the reader should note that if class intervals are given like 0-10, or 10-20, it is always presumed that the upper limit is exclusive i.e. an observation exactly to the upper limit is not included in that class.

b. Inclusive method:

Under the “Inclusive method” of classification, the upper limit of one class is included in that class itself.

Income (Tk.)	No of Employees
800-899	50
900-999	100
1000-1099	200

In the class 800-899 we include persons whose income is between Tk 800 and Tk. 899. If the income of persons is exactly Tk. 900 he is included in the next class.

➤ **Principles of Classification:**

It is difficult to lay down any hard and fast rules for classifying the data as the type of classification.

1. The number of classes should preferably be between 5 and 15. However, there is no rigidity about it. The classes can be more than 15 depending upon the total number of observations in the series and the details required, but they should not be less than five because in that case, the classification may not reveal the essential characteristics.

Sturges suggested the following formula for determining the approximate number of classes:

$$K = 1 + 3.322 \log N$$

K= The approximate number of classes

N= Total number of observations

Log= The ordinary logarithm to the base of 10.

However, the precise number of classes to be used for a given variable depends upon personal judgment and other considerations such as the details required, The case of calculation of further statistical work, etc.

Example-1:

The profits (in lakhs of Tk's) of 30 Bangladeshi companies for the year 2005-2006 are given below:

18	16	23	37	35	49	63	65	55	45
58	57	69	20	22	35	42	37	42	48
53	49	65	39	48	67	25	29	58	65

Classify the above data taking a suitable class interval.

Solution:

Let us determine the suitable class interval with the help of the following formula:

Class interval, $i = \frac{Range}{K}$ (nearest integer number)

We have, $N=30$,

Highest value=69,

lowest value=16

Range = Highest value - lowest value

Range = $69 - 16 = 53$

Where, the approximate number of class $K=1+3.322\log N$

$K=1+3.322\log 30=5.91 \approx 6$,

$$i = \frac{Range}{K} = \frac{53}{5.91} = 8.97 \text{ or } 9$$

We will take 10 as the class interval and the first class is 15-25.

Frequency Distribution of the profits:

Profit (Tk.lakhs)	Tally	No. of companies
15-25		5
25-35		2
35-45		7
45-55	I	6
55-65		5
65-75		5
		Total 30

Problem-2:

The following are the marks of the 30 students in statistics. Prepare a frequency distribution taking a suitable class interval.

12 33 23 25 18 35 37 49 54 51 37 15

27 33 42 45 47 55 69 65 63 46 29 18

37 45 46 59 29 55

Problem-3:

Form a frequency distribution for the following data-

30, 42, 30, 54, 40, 48, 15, 17, 51, 42, 25, 41, 30, 27, 42, 36, 28, 26, 37, 54, 44, 31, 36, 40, 36, 22, 30, 31, 19, 48, 16, 42, 32, 21, 22, 40, 33, 41, 21.

Problem-4:

Form a frequency distribution for the following data-

67, 34, 36, 48, 49, 31, 61, 34, 43, 45, 38, 32, 27, 61, 29, 47, 36, 50, 46, 30, 46, 32, 30, 33, 45, 49, 48, 41, 53, 36, 37, 37, 47, 62, 46, 50, 28, 35, 35, 38, 36, 46, 43, 34, 62, 69, 50, 28, 44, 43, 60, 39.

Problem-5:

Form a frequency distribution for the following data:

37, 39, 58, 45, 47, 69, 76, 58, 41, 46, 56, 64, 67, 71, 75, 43, 37, 39, 58, 45, 47, 69, 76, 58, 41, 46, 53, 63, 67, 71, 76, 43, 37, 39, 58, 45, 47, 60, 76, 58, 41, 46, 54, 64, 67, 71, 75, 43, 37, 39, 58, 45, 47, 69, 76, 58, 41, 46, 56, 52.

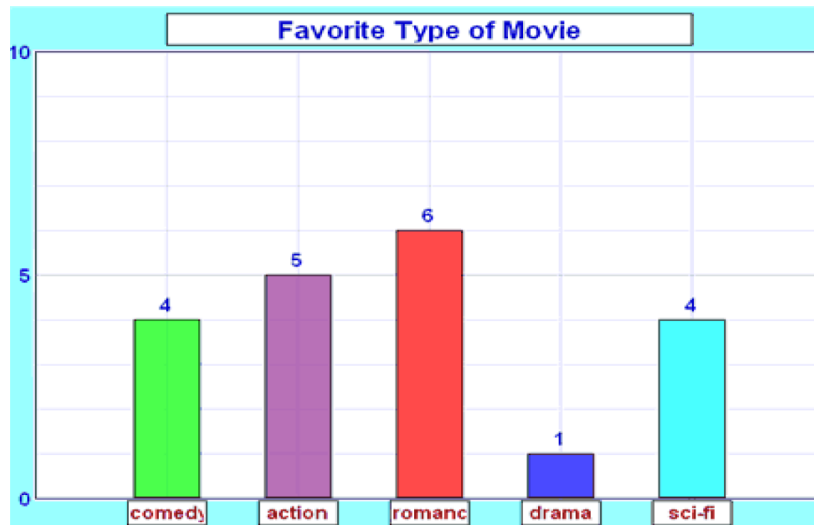
Simple Bar Diagrams:

A simple bar diagram is used to represent only one variable. For example, the figures of sales, production, population etc, for various years may be shown by means of a simple bar diagram. However, an important limitation of such diagrams is that they can present only one classification or one category of data.

A Bar Graph (also called Bar Chart) is a graphical display of data using bars of different heights.

Imagine you just did a survey of your friends to find which kind of movie they liked best.

<i>Table: Favorite Type of Movie</i>				
Comedy	Action	Romance	Drama	Science Fiction
4	5	6	1	4



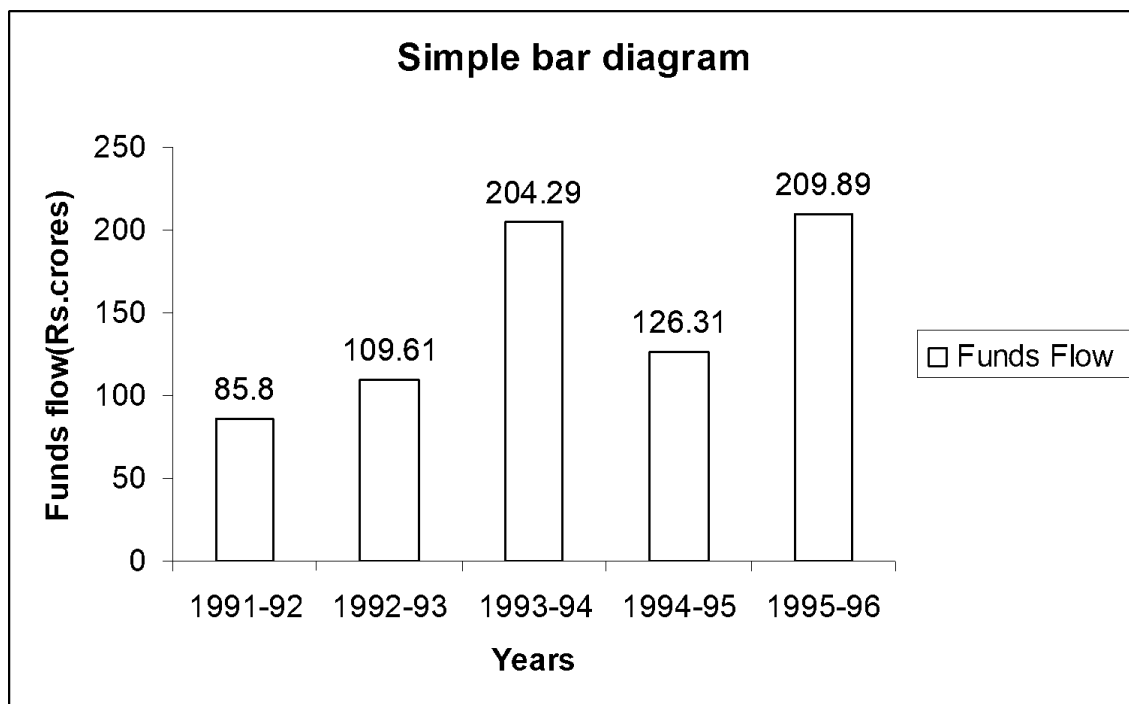
It is a really good way to show relative sizes: we can see which types of movies are most liked, and which are least liked, at a glance.

Example-6:

The funds flow of Goodwill India Ltd from 1991-92 to 1995-96 are given below:

Year	Funds Flow (Rs.Crores)
1991-92	85.80
1992-93	109.61
1993-94	204.29
1994-95	126.31
1995-96	209.89

Represent this data by a suitable bar diagram.

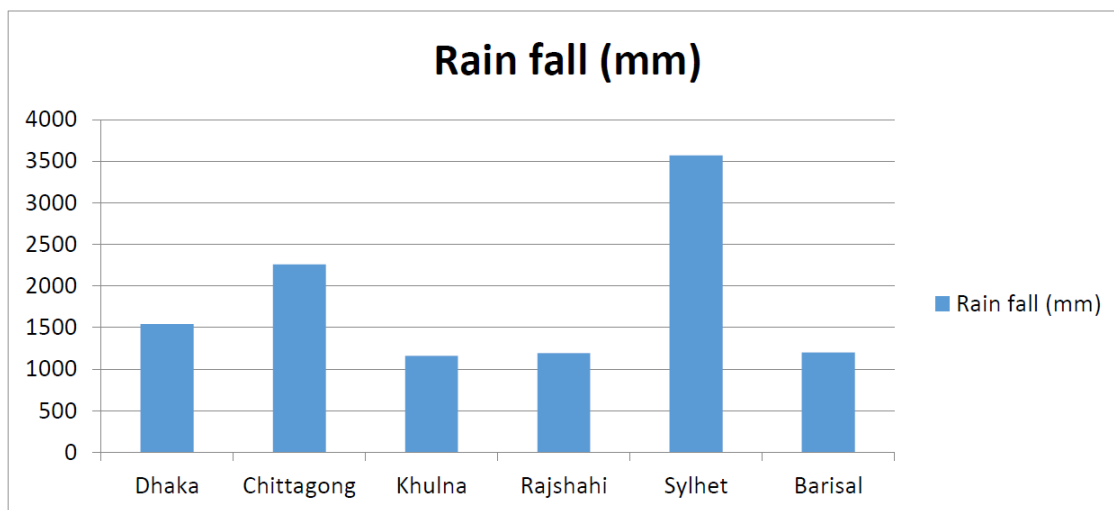


Example-7:

Draw the Bar Diagram from the following data-

Division	Rain fall (mm)
Dhaka	1540
Chittagong	2260
Khulna	1159
Rajshahi	1192
Sylhet	3568
Barisal	1200

Solution:

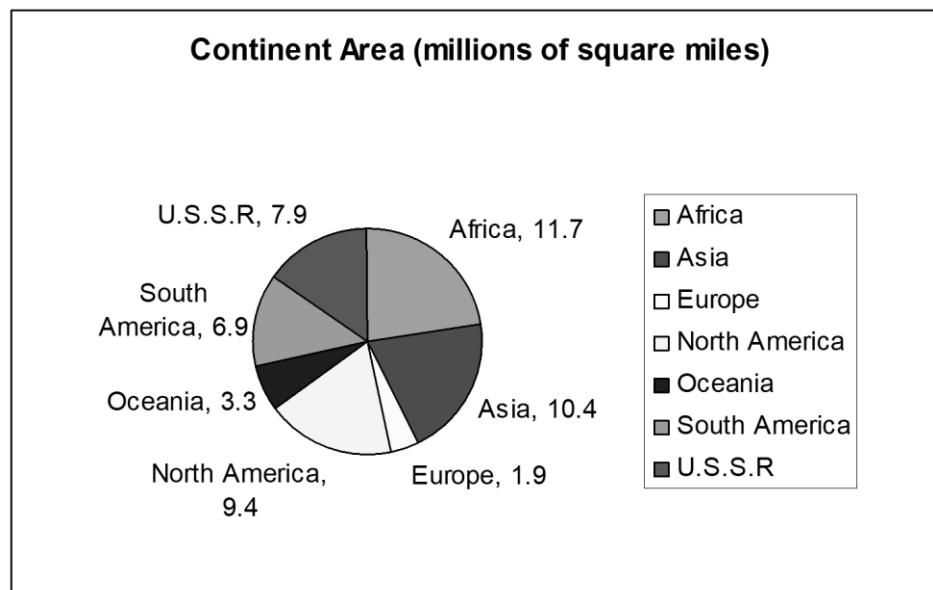


Example-8:

Areas of continents of the world

Continent	Area (millions of square miles)
Africa	11.7
Asia	10.4
Europe	1.9
North America	9.4
Oceania	3.3
South America	6.9
U.S.S.R	7.9
Total	51.5

Draw a pie chart from the above data.



To construct such a diagram we use the fact that “the whole” (51.5 in the above illustration) corresponds to the total number of degrees in the circular arc, namely 360° . This 360° is then proportionately divided among the various components of the whole. Thus the above illustration; the arc of the segment representing

Asia subtends on an angle of $73^\circ (= \frac{360^\circ}{51.5} \times 10.4)$ at the center of the circle.

This diagram should be sparingly used, especially if there are many segments.

Pie chart

In order to draw a **pie chart**:

- 1 Calculate the angle for each category.
- 2 Draw a circle, mark the centre and draw a radius.
- 3 Measure and draw the angle for the first category.
- 4 Measure and draw the angle for each further category, in order.
- 5 Add data labels / complete a key.



Problem-9:

Brian asks 60 people what their favorite color is and separates the answers into 5 categories. His results are shown in the table below. Draw a **pie chart** to display Brian's results.

Colour	Red	Blue	Green	Yellow	Other
Frequency	10	13	24	5	8

To draw the pie chart, we need to calculate the size of the angle needed for each category using the formula.

$$\text{'Red' angle} = \frac{10}{60} \times 360 = 60^\circ$$

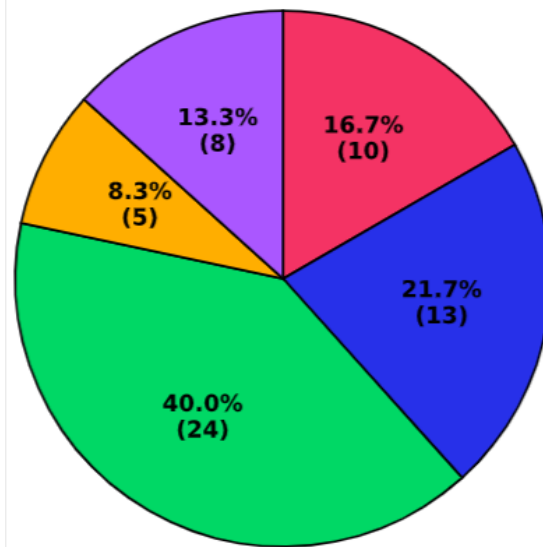
$$\text{'Blue' angle} = \frac{13}{60} \times 360 = 78^\circ$$

$$\text{'Green' angle} = \frac{24}{60} \times 360 = 144^\circ$$

$$\text{'Yellow' angle} = \frac{5}{60} \times 360 = 30^\circ$$

$$\text{'Other' angle} = \frac{8}{60} \times 360 = 48^\circ$$

Now we have all the angles, we can draw the chart.



Problem-10:

Mr. X asked his 20 friends which types of movies they liked very much. The following data was found from his survey.

<i>Table: Favorite Type of Movie</i>				
Comedy	Action	Romance	Drama	Sci Fi
4	5	6	1	4

Show the above data with a pie chart.

It is a really good way to show relative sizes: it is easy to see which movie types are most liked, and which are least liked, at a glance.

First, put your data into a table (like above), then add up all the values to get a total:

<i>Table: Favorite Type of Movie</i>					
Comedy	Action	Romance	Drama	Sci Fi	TOTAL
4	5	6	1	4	20

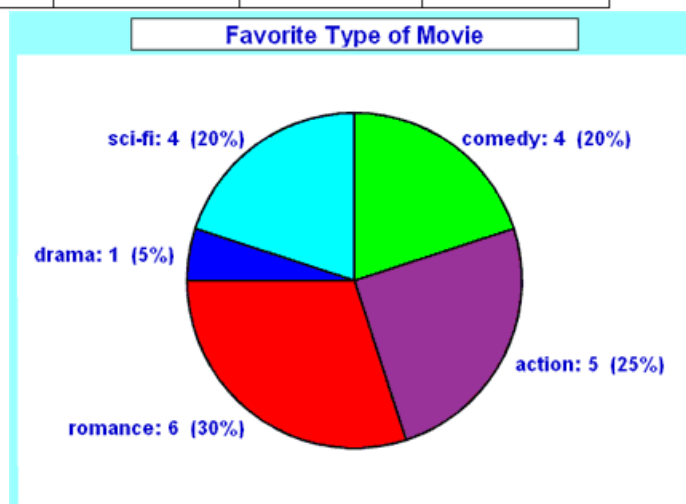
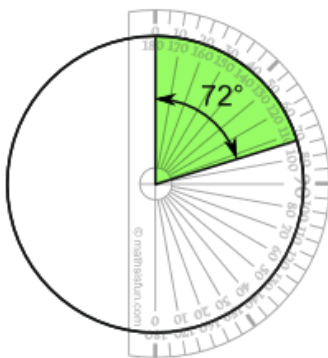
Next, divide each value by the total and multiply by 100 to get a percent:

Comedy	Action	Romance	Drama	SciFi	TOTAL
4	5	6	1	4	20
$\frac{4}{20}$ 20%	$\frac{5}{20}$ =25%	$\frac{6}{20}$ =30%	$\frac{1}{20}$ =5%	$\frac{4}{20}$ =20%	100%

Now to figure out how many degrees for each "pie slice" (correctly called a sector).

A Full Circle has **360 degrees**, so we do this calculation:

Comedy	Action	Romance	Drama	SciFi	TOTAL
4	5	6	1	4	20
20%	25%	30%	5%	20%	100%
$\frac{4}{20} \times 360^\circ$ = 72°	$\frac{5}{20} \times 360^\circ$ = 90°	$\frac{6}{20} \times 360^\circ$ = 108°	$\frac{1}{20} \times 360^\circ$ = 18°	$\frac{4}{20} \times 360^\circ$ = 72°	360°



Sub-divided or component Bar Diagrams:

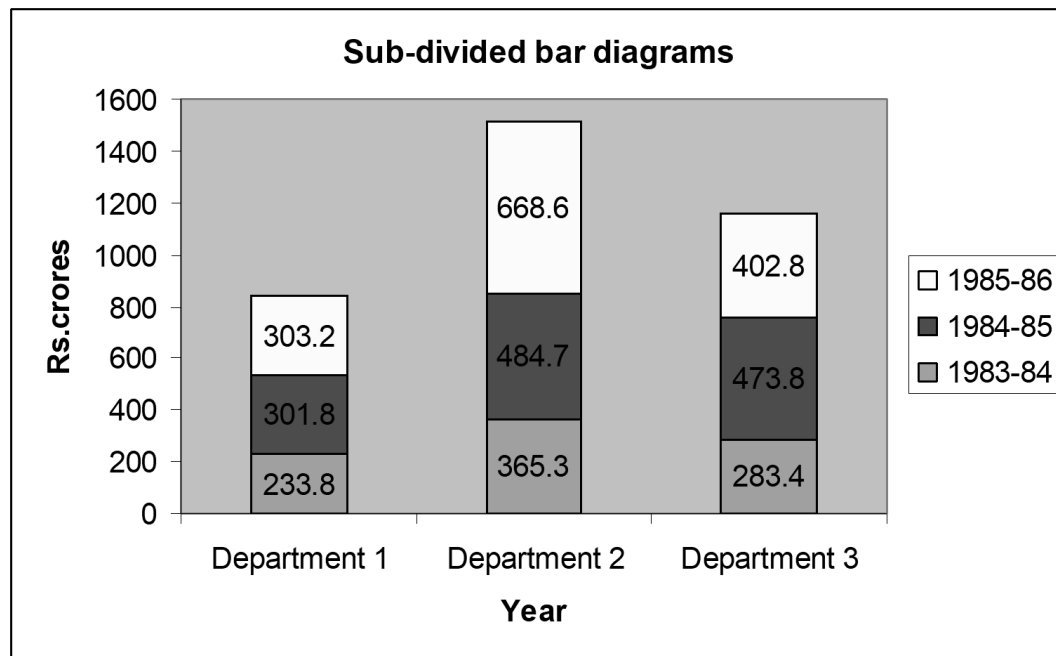
These diagrams are used to represent various parts of the total. For example, the number of employees in various departments of a company may be represented by sub-divided bar diagrams. While constructing such a diagram the various components in each bar should be kept in the same order. To distinguish between the different components, it is useful to use different shades or colors. Sub-divided bar diagrams can be vertical as well as horizontal.

Example-11:

Represent the following data by sub-divided bar diagrams

(in Rs.Crores)

Year	Department 1	Department 2	Department 3
1983-84	233.8	365.3	283.4
1984-85	301.8	484.7	473.8
1985-86	303.2	668.6	402.8



Multiple bar Diagrams:

In a multiple-bar diagram, two or more sets of interrelated data are represented. The technique of drawing such a diagram is the same as that of a simple bar diagram. The only difference is that since more than one phenomenon is represented, different shades, colors, or crossings are used to distinguish between the bars.

Example-12:

	Corporate Sector Profits(Rs.crores)	
	1994-95	1995-96
Gross profits	3104	3123
Profits before tax	1663	1376
Profits after tax	1219	982

Graphs of Frequency Distributions:

A frequency distribution can be presented graphically in any of the following diagrams:

1. **Histogram**
2. Frequency Polygon
3. Smoothed frequency curve
4. **Cumulative frequency curves or ‘Ogives’.**

Construction of Histogram When Class-intervals are Equal:

When class intervals are equal, take the frequency on the Y-axis, and the variable on the X-axis and construct adjacent rectangles. In such a case the heights of the rectangles will be proportional to the frequencies.

Example-13:

Represent the following data by a histogram:

Size Class	Frequency	Size Class	Frequency
0-10	5	50-60	10
10-20	11	60-70	8
20-30	19	70-80	6
30-40	21	80-90	3
40-50	16	90-100	1

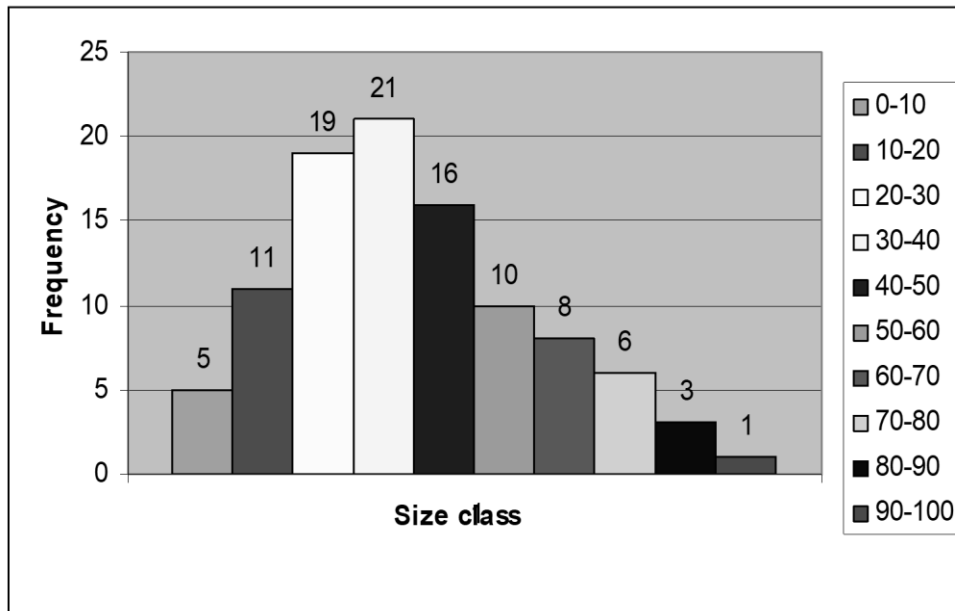


Fig: Histogram when Class intervals are equal

Problem-14:

Draw the histogram from the following data-

Class Interval	Frequency
21 - 25	2
26 - 30	4
31 - 35	3
36 - 40	2
41 - 45	5

Problem-15:

Draw the histogram from the following data-

Class Interval	Frequency
14.5 - 19.5	4
19.5 - 24.5	4
24.5 - 29.5	4
29.5 - 34.5	8
34.5 - 39.5	4
39.5 - 44.5	10
44.5 - 49.5	2
49.5 - 54.5	3